

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC HOMEWORK & ASSESSMENT SUBMISSION

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Course Code & Title:	NPTEL-CS06 - PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON
Task & Channel:	NPTEL Week 5 Assignment: Heaps & Priority Queues [NPTEL]

Assignment Description:

Weekly graded assessment for NPTEL DSA course: 5 MCQs on Min-Heap, Max-Heap, Heapify complexity, and Median maintenance.

Technical Solution:

NPTEL Week 5 Assessment Solutions

Q1: What is the time complexity to build a heap of n elements from an unsorted array?

- **Correct Answer:** $O(n)$

- **Mathematical Rationale:** Building a heap bottom-up via Floyd's build-heap algorithm takes linear time $O(n)$ because the sum of heights $\sum(h/2^h)$ converges to a constant.

Q2: In a max-heap with 10 elements, what is the minimum number of comparisons needed to find the minimum element?

- **Correct Answer:** 5

- **Mathematical Rationale:** In a max-heap of 10 elements, the minimum element must reside in one of the leaf nodes (indices $\text{floor}(n/2)$ to $n-1$, i.e., indices 5 to 9 = 5 leaves). Finding the minimum among 5 leaves requires 4 comparisons.